

Voice Command Controlled Drone

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With path-breaking progress in UAV technology, drones are now being readily adopted and deployed for several uses. But despite the advancement, many physically challenged people are unable to operate them due to certain limitations in maintaining

drone stability in the air. Now they too can fly a drone with the help of voice commands. And the best part is that the voice commands can be given in English or any other language.

For this fascinating project, the author used Raspberry Pi with AIY Voice Bonnet as a ground control sta-

tion (GCS). You can build the GCS with any Linux PC as well. This GCS (or PC) uses voice commands to control the Tello drone. Once the drone system recognises the voice command, a connection is established with which the drone can be controlled.

Tello drone used by the author is shown in Fig. 1. The AIY Voice Bonnet mounted on top of the Raspberry Pi board is shown in Fig. 2.

Prerequisite

Use of Raspbian based Raspberry Pi is recommended as it is portable, small, and can be easily used for controlling the drone. Install the following modules:

- Speech_recognition
- PyAudio
- TelloPython

The Speech_recognition and PyAudio modules help capture the voice and recognise the voice commands. TelloPython is used for establishing connection with the Tello drone. It also allows drone control and video capture.

TelloPython facilitates the following functions:

- Drone takeoff using `drone.take-off()`
- Drone navigation in the forward direction using `drone.forward(cm)`
- Rotating the drone to 90 degrees in clockwise or anti-clockwise direction using `drone.cw(90)`
- Forward, backward, left or right mid-air flips using `drone.flip('l')`
- Live stream a video from the drone camera to the PC using `drone.streamon()`
- Land the drone using `drone.land()`

Coding

The AIY Voice Bonnet with speech recognition module allows capture and recognition of voice commands. To enable it, you need to import two



Fig. 1: Tello drone



Fig. 2: AIY Voice Bonnet attached to Raspberry Pi